Form B

1. (a) and (c)

2. (b) and (e)

4. (a) 68%

   (b) $np \geq 10$ and $n(1-p) \geq 10$.

   (c) It is reasonable so long as the ratio: $\dfrac{\text{population size}}{\text{Sample size}} \geq 20$.

5. (e) qf-qq(~differences)

   (f) The dots do not perfectly align with a line, but they mostly do, making it appear that the population of differences

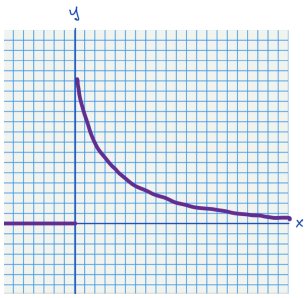
   glaucoma − healthy

   is somewhat like a normal distribution.

   (g) The 90% CI for μ_{diff} is

   $$-1.375 \pm 1.89458 \frac{2.50357}{\sqrt{8}} \quad \text{or} \quad (-3.052, 0.302)$$

6. (a)



   (b) qexp(0.83, rate = 0.2)

7. (a) The sample proportion has $SE_{\hat{p}} = \sqrt{\dfrac{p(1-p)}{n}}$, and a test (z) statistic:

   $$\hat{p} = \frac{49}{106} = 0.4623 \quad \Rightarrow \quad z = \frac{0.4623 - 0.38}{\sqrt{\dfrac{(0.38)(0.62)}{106}}} \doteq 1.745$$

   (b) 1-pnorm(1.745) (right-tailed)

   (c) Letter (ii)

   (d) The result is statistically significant at the 6% level. We reject that $p = 0.38$ in favor of the alternative, that $p > 0.38$.

8. (a) The samples are "independent" and "random," which we need. Though sample sizes (21 and 24) are close to 30, we have no knowledge of the population to assure us that $\bar{x}_1, \bar{x}_2$ are normal, and thereby allowing small sample sizes.

(b) $qt(0.98, df = 42.5869)$

(c) Take μ_1 = mean hemoglobin level for breast-fed infants

μ_2 = mean hemoglobin level for formula-fed infants

$$\bar{x}_1 - \bar{x}_2 \pm t_c \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = (11.2 - 10.5) \pm (2.119)\sqrt{\frac{1.5^2}{21} + \frac{1.9^2}{24}}$$

or $(-0.375, 1.775)$